An Explainable AI Framework for Diabetic Foot Ulcer Detection and Personalized Footwear Recommendation

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***Abstract*—Diabetic Foot Ulcers (DFUs) are a major complication of diabetes, often leading to infections, hospitalizations, and amputations if not detected early. This study proposes an integrated deep learning and IoT-based system for the early detection, classification, and prevention of DFUs. An EfficientNetB0 model is utilized to classify DFU images with high accuracy, while plantar foot type classification is performed using WaveNet, ResNet, and an Attention Mechanism to analyze plantar pressure data. For personalized footwear recommendation, a Decision Tree model is employed, enhanced with Explainable AI techniques (SHAP, LIME, and Grad-CAM) to provide transparency in decision-making. The system captures real-time foot health parameters using multiple IoT sensors (pressure, temperature, heart rate, gyroscope) embedded in smart footwear, enabling continuous monitoring and risk prediction. The developed models show promising results in improving early DFU diagnosis, personalized prevention strategies, and clinical decision support. Future enhancements aim to scale the solution with larger datasets, real-time cloud integration, and clinical validation. This research provides a novel, explainable, and wearable AI-driven framework for improving diabetic foot care outcomes.**

Keywords—Explainable AI, Diabetic Foot Ulcer, Recommendation System, Deep Learning, Neural Network, IoT

I. Introduction

Diabetic Foot Ulcer (DFU) detection and classification are crucial for preventing complications such as infections and amputations. Traditional manual diagnosis is often subjective and time-consuming. A pre-trained EfficientNetB0 model is fine-tuned with transfer learning to accurately classify foot ulcer images. Global Average Pooling (GAP) is used to extract high-level features, maintaining a lightweight architecture suitable for deployment.

In addition to ulcer detection, our framework includes a plantar foot type classification module, utilizing a hybrid model that fuses WaveNet, ResNet, and Attention mechanisms to analyze plantar pressure data and foot shape images. This model captures temporal and spatial features for classifying foot types—flat foot, high arch, and normal arch—offering essential biomechanical insights.

To enhance clinical relevance and transparency, the framework incorporates Explainable AI tools such as SHAP, LIME, and Grad-CAM. These methods provide interpretability in model predictions, boosting trust and supporting informed decision-making. Data augmentation strategies, including image rotation, flipping, and noise addition, were employed to improve model generalization. A Conditional Generative Adversarial Network (cGAN) further addresses dataset limitations by synthesizing plantar pressure data for robust foot type classification.

The system also integrates a real-time monitoring solution using IoT sensors embedded in footwear. These sensors track pressure, temperature, heart rate, and gyroscopic movement, enabling continuous health surveillance and timely risk detection. Sensor data, combined with clinical information, feeds into an AI-driven personalized footwear recommendation engine. Unlike static rule-based systems, our model adapts dynamically to foot health variations, offering footwear suggestions that account for ulcer severity, foot type, and plantar pressure distribution.

Patients with flat feet are guided toward arch-support footwear; those with high arches receive cushioned shoes, while normal-footed individuals are recommended balanced orthopedic options. Real-time alerts are delivered through a connected mobile application, ensuring proactive care.

This work presents an integrated pipeline that spans DFU detection, foot classification, risk assessment, and intelligent intervention. The dual-module system ensures a continuous feedback loop where updated assessments inform dynamic footwear recommendations, transforming reactive treatment into proactive care.

The proposed solution advances the current state-of-the-art by fusing deep learning, IoT, and XAI into a unified framework, targeting early diagnosis, personalized prevention, and improved mobility. The final outcome is a deployable smart wearable prototype, designed to reduce ulcer recurrence and improve the quality of life for diabetic patients, while supporting clinical practitioners with transparent, reliable insights.

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II. LITERATURE SURVEY

Recent innovations in diabetic foot care focus on deep learning, IoT-enabled sensing, and explainable AI to detect and manage diabetic foot ulcers (DFUs). Deep learning models like CNNs, EfficientNet, and ResNet have shown high accuracy in DFU classification by extracting features from clinical images (Biswas et al., 2023 [5]; Nagaraju et al., 2023 [10] ; Toofanee et al., 2023 [13]). Multi-scale fusion techniques and explainable models such as those by Goyal et al. (2018) [17], Das et al. (2022) [15], and Biswas et al. (2024) [1] further enhance diagnostic reliability.

Smart insoles equipped with pressure sensors are widely used to monitor plantar pressure, aiding early identification of ulcer risks. Studies using statistical and machine learning techniques for pressure data classification (Agrawal et al., 2024 [3]; Girão et al. [7], 2023; Jia et al., 2024 [8]) have evolved into deep learning models predicting pressure dynamics (Zhang et al., 2024 [14]; Li et al., 2024 [23]). For foot type classification, deep learning models like WaveNet and ResNet have outperformed traditional methods (Han et al., 2023 [3]; Zhao et al., 2023 [4]). Recent advances incorporate thermal imaging and sensor fusion (Zhang et al., 2024 [14]; Wu et al., 2024 [22]) for improved accuracy.

Explainable AI tools such as SHAP, LIME, and Grad-CAM have been integrated into DFU detection to provide interpretability (Saraswat et al., 2022 [12]; Biswas et al., 2024 [30]; Reyes-Luévano et al., 2023 [11]; Alzubaidi et al., 2020 [16]; Yap et al., 2021 [26]; Iman et al., 2023 [28]), boosting clinical trust. Footwear interventions, traditionally static, are increasingly leveraging AI and real-time sensor data to deliver dynamic recommendations (Wang et al., 2020 [24]; Zhang et al., 2024 [14]; Li et al., 2024 [23]).

However, current systems face significant limitations. Most lack integration of image and sensor data, operate passively without real-time alerts, rely on opaque models with poor explainability, and offer generic footwear suggestions. Manual inspection remains a bottleneck, and personalization is minimal. Similar works have been carried out in [31-33].

The proposed system overcomes these challenges through a comprehensive AI-IoT platform integrating EfficientNetB0-based DFU classification, hybrid WaveNet-ResNet foot type analysis, and real-time plantar monitoring. It delivers automated alerts via a mobile app, improves decision transparency using SHAP, LIME, and Grad-CAM, and tailors footwear recommendations using a multi-input strategy combining DFU risk, foot type, and live sensor feedback. Continuous model retraining enables adaptive care over time. With mobile integration, the system supports remote monitoring and telemedicine, offering intelligent, timely, and personalized interventions for diabetic foot care.

III. PROPOSED SYSTEM

The proposed framework is an integrated Deep Learning and Explainable AI (XAI)-enabled system aimed at early detection, classification, prevention, and continuous monitoring of Diabetic Foot Ulcers (DFUs). By combining foot image analysis, plantar pressure classification, personalized footwear recommendations, and IoT-enabled real-time monitoring, the system offers a holistic and intelligent approach to diabetic foot care.

A. System Architecture Overview

The architecture (Figure 1) begins with data acquisition from foot images and plantar pressure sensors embedded in smart footwear. The DFU detection pipeline employs a fine-tuned EfficientNetB0 model, while foot type classification leverages a hybrid deep learning network combining WaveNet, ResNet, and an attention mechanism. These classification outputs feed into a decision-making module, which generates personalized footwear recommendations. To ensure transparency, XAI tools like SHAP, LIME, and Grad-CAM are integrated for visual and feature-based interpretability. The final output is displayed via a user-friendly mobile application for real-time feedback and monitoring, with optional cloud storage for analysis and updates.

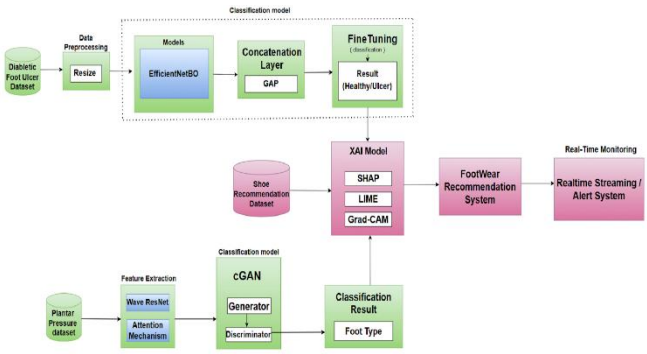

Figure 1: Overall Architecture Diagram

B. DFU Image Classification

Foot images, captured via smartphone or clinical cameras, undergo preprocessing (resizing, normalization) and are classified as Healthy or Ulcer using the EfficientNetB0 model. This process allows for early ulcer detection and intervention, reducing the risk of complications and enabling patient-specific treatment plans(Figure 2).

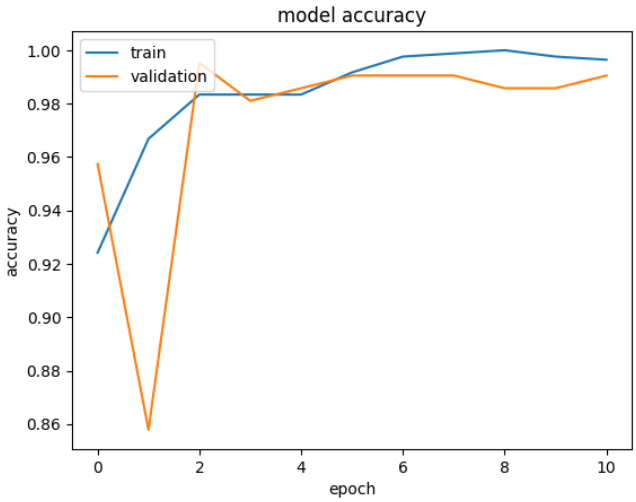

Figure 2 Model Accuracy of DFU

C. Plantar Pressure Classification

Real-time plantar pressure data is gathered from smart insoles. This data is fed into a hybrid deep learning model (WaveNet + ResNet + Attention) to classify foot types into Flat Foot, Normal Arch, or High Arch. Identifying foot structure helps uncover biomechanical stress zones,

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improving the system's ability to predict ulcer-prone areas and guide preventive care (Figure 3).

D. *Explainable AI (XAI) Integration*

To enhance clinical trust, XAI methods such as SHAP, LIME, and Grad-CAM are applied to both image and sensor-based models. These tools offer interpretable visualizations of model decisions, allowing healthcare providers to understand the basis for ulcer detection and foot type classification, promoting transparency and accountability in AI-driven healthcare.

E. *Personalized Footwear Recommendation*

The system recommends footwear based on combined analysis of DFU presence and foot type. Custom recommendations include features like pressure offloading zones, arch support, and enhanced cushioning. This personalization reduces ulcer recurrence risk and improves patient comfort and adherence.

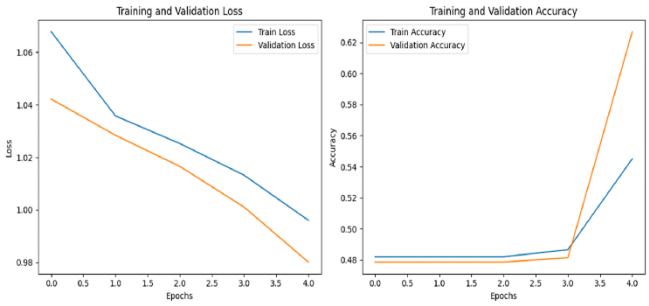

Figure 3: Training and validation Accuracy of Plantar Pressure

F. *Real-Time Monitoring and Alert*

Smart footwear continuously monitors metrics such as pressure shifts, gait, and temperature. Abnormal patterns trigger instant alerts via the mobile app, enabling early warnings and proactive intervention. This component ensures consistent tracking of foot health, especially in high-risk patients.

This modular, scalable framework integrates AI-driven diagnostics with patient-specific recommendations and explainable insights, making it a powerful solution for preventing and managing diabetic foot complications.

IV. RESULTS AND DISCUSSION

This section presents the performance evaluation of the models used for diabetic foot ulcer (DFU) detection using a combination of standard classification metrics(Table I and Table II). The metrics employed include Accuracy, Precision, Recall (Sensitivity), F1 Score, and advanced explainability methods. Each metric is explained with its mathematical formulation and followed by an interpretation of the observed results for the EfficientNetB0 and CGAN models.

Table I Evaluation Metrics for EfficientNetB0

Class	**Precision**	**Recall / Sensitivity**	**F1 Score**
Abnormal (Ulcer)	0.9792	0.9592	0.9691
Normal (Healthy skin)	0.9608	0.9800	0.9703
Overall Accuracy	0.9697	0.9697	0.9697
Macro Avg	0.9700	0.9696	0.9697
Weighted Avg	0.9700	0.9696	0.9697

A. *Accuracy*

Accuracy is the proportion of total correctly classified instances out of all predictions made. It is calculated using the formula (1) is as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (1)$$

EfficientNetB0 achieved a high accuracy of 96.97%, demonstrating its robustness in classifying both ulcer and non-ulcer images with minimal error. In contrast, the CGAN model achieved a lower accuracy of 66.97%, indicating its limitations in effectively learning discriminative features for DFU classification.

Table II Evaluation Metrics for CGAN

Class	**Precision**	**Recall (Sensitivity)**	**F1 Score**
Abnormal (Ulcer)	0.6792	0.6592	0.6691
Normal (Healthy skin	0.6608	0.6800	0.6703
Accuracy	0.6697	0.6697	0.6697
Macro Average	0.6700	0.6696	0.6697
Weighted Average	0.6701	0.6697	0.6698

B. *Precision*

Precision measures the model's ability to correctly identify only the relevant (ulcer) instances among all positively predicted cases. The formula (2) is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

A high precision means fewer false positives. For the ulcer class, EfficientNetB0 achieved a precision of 0.9792, indicating that most of the instances it predicted as ulcers were actually correct. On the other hand, CGAN reached a precision of 0.6792, showing a higher tendency to misclassify normal feet as ulcers.

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C. Recall (Sensitivity)

Recall or Sensitivity quantifies the model's ability to correctly detect all actual positive cases. It is given by (3)

$$Recall = \frac{\text{TP}}{\text{TP} + \text{FN}} \qquad (3)$$

EfficientNetB0 exhibited a recall of 0.9592 for the ulcer class, confirming its effectiveness in identifying the majority of ulcer cases and reducing the number of missed diagnoses. The CGAN model, however, showed a lower recall of 0.6592, implying it failed to identify several true ulcer cases.

D. F1 Score

The formula (4) is as follows:

$$\text{F1 Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \qquad (4)$$

EfficientNetB0 reported an F1 score of 0.9691 for the ulcer class and 0.9703 for the normal class, which confirms its balanced performance across both classes. In contrast, CGAN produced an F1 score of 0.6691 for ulcers and 0.6703 for normal feet, reflecting moderate effectiveness.

E. Confusion Matrix

For EfficientNetB0, the matrix shows minimal misclassifications, validating the model's high precision and recall. The CGAN model's matrix reveals more false positives and false negatives, consistent with its lower overall metric scores.

F. Explainability Metrics

To support clinical adoption, explainable AI tools were integrated. SHAP provided feature-level insights into model predictions, LIME explained individual predictions locally, and Grad-CAM visualized the regions of interest in the foot images that influenced decisions. These methods collectively enhanced the transparency and trustworthiness of the system's recommendations.

In summary, the EfficientNetB0 model consistently outperformed CGAN in all evaluated metrics, achieving near-perfect classification performance suitable for clinical use. The inclusion of interpretability methods further ensures that the system is not only accurate but also explainable and reliable for real-world diabetic foot ulcer management.

V. CONCLUSION AND FUTURE WORK

This study presents a deep learning and IoT-based system for the early detection, classification, and prevention of Diabetic Foot Ulcers (DFUs). The system operates in three stages: EfficientNetB0 is used for accurate DFU detection and classification; WaveNet, ResNet, and attention-based ResNet models are employed for plantar foot type classification using pressure data; and a Decision Tree model, enhanced with SHAP, LIME, and Grad-CAM, is used to recommend personalized footwear based on multiple sensor readings. Overall, the system not only identifies DFUs but also supports proactive intervention and prevention, offering a practical solution for diabetic foot management.

Future work will focus on expanding the dataset through collaborations with healthcare institutions, enhancing model generalization. Developing a cloud-based system for real-time monitoring and remote access will improve clinical usability. Exploring advanced models like reinforcement learning and Graph Neural Networks (GNNs) could enable more adaptive recommendations. Combining multimodal sensor data into a unified model may further improve prediction accuracy. Clinical validation through trials and sensor miniaturization will increase real-world effectiveness and user comfort. Integration with telemedicine platforms will support remote care, making the system more accessible and scalable.

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